PEDRO HENRIQUE ROCHA DE ANDRADE

Software developer · Python, Node.js, data and *self-hosted* infrastructure

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SUMMARY

Computer Engineering student (IFF) who builds and runs complete systems, from database to deployment: academic and commercial web applications, a *self-hosted* LaTeX platform and GPU-powered scientific data *pipelines*, always with tests, automation and documentation.

FEATURED PROJECTS

ReLaTeX

github.com/pedroiff0/relatex

Creator and maintainer · 2025 – present

</> Docker, Node.js, MongoDB, Redis, Pug, Nginx, TeX Live

- *Self-hosted* LaTeX platform (Overleaf CE fork) with IFF classes following ABNT standards, OLED theme, WCAG AA accessibility and export to DOCX and PPTX.
- *Agent Gateway*: API for AI agents with per-user tokens, LGPD consent, rate limiting and human approval of every edit, project creation or organisation proposal, with undo.

Verdementa retail suite

private repositories

Developer · retail systems

</> Python, Flask, SQLAlchemy, Alembic, AdminLTE, CI

- **Cash desk**: daily cash closing (sales, deposits, vouchers and drawer count), printable reports, REST endpoints and versioned *migrations* in CI.
- **Stock and Spreadsheet tool**: inventory, purchases and NFC-e invoice XML turned into automatic reports.

Assessment platform

private repositories

Creator · three products on one core

</> Node.js, Express, MongoDB, EJS, LaTeX, Expo

- Reusable question bank that generates LaTeX exams with multiple variants (random selection and order).
- Editions for teachers (shared bank) and civil-service exam candidates (study tracks by exam, mock tests and an Expo mobile app with feature parity).

Academic System

private repository

Creator · IFF students

</> Node.js, Express, MongoDB, EJS, JWT, Docker

- Curriculum, class log, grades, attendance, transcript and GPA, multi-programme (Computer Engineering and Information Systems), with a bot API.

Finance App

private repository

Creator · personal finance

</> Node.js 20, Express, MongoDB, Zod, Jest, Docker, Helmet

- Envelope budgeting, goals and a multi-broker investment portfolio, with automated tests.

Spectral data pipeline and web viewer

SAB 2026

Research · PIBIC/CNPq

</> Python, scikit-learn, CUDA, HDBSCAN, t-SNE

- Chunked GPU processing of about 900 thousand GALAH DR4 spectra and an interactive web viewer presented at the 2026 SAB Annual Meeting.

STACK

Languages

Python · JavaScript / Node.js · SQL · C++ · Java · LaTeX

Web and data

Express · Flask · MongoDB · Redis · scikit-learn · CUDA

Infrastructure

Docker · Linux · Nginx · Tailscale · Git and CI · 24/7 home server

HOW I WORK

- Tests and CI before every release.
- Documentation next to the code (README, CHANGELOG, decisions).
- Secure by default: per-user tokens, LGPD, dependency review.
- AI agents with human approval and an audit trail.

HIGHLIGHTS

1st place, Exact Sciences
XVI MCTIA, 2025

3rd place, Exact Sciences
IX FEBIC, 2024

Organiser, IV SCEG
IFF, 2024: workshops and *hackathon*

EDUCATION

Computer Engineering (B.Eng.)
IFF · 2023 – present

Technical Diploma in Informatics
IFF · 2020 – 2022

Cisco Networking Academy
Networking, operating systems and server administration

LANGUAGES

Portuguese ●●●●●

English ●●●●○

Spanish ●●○○○

Updated November 3, 2026. Academic CV at phrandrade.com.